

# **White Paper: Decentralized Identity & Reputation Tracker on OPNet**

## **Abstract**

This paper introduces a decentralized identity and reputation system built on **Bitcoin Layer 1 via OPNet**. The solution enables users to establish verifiable digital identities tied to Bitcoin addresses and build trust through transparent, immutable reputation scores. It is designed for freelancers, marketplaces, and communities where trust is critical.

## **Problem Statement**

- Current reputation systems are centralized (e.g., Upwork, Fiverr, Airbnb).
- Users cannot port their reputation across platforms.
- Centralized control leads to censorship, manipulation, and lack of transparency.

## **Proposed Solution**

- A decentralized identity layer linked to Bitcoin addresses.
- Smart contracts record verifiable actions (transactions, reviews, contributions).
- Reputation scores are calculated transparently and stored immutably.
- Portable trust: users can carry their reputation across platforms.

## **Key Features**

- **Identity Binding:** Users register a Bitcoin address as their identity.
- **Reputation Ledger:** Immutable record of ratings, reviews, and contributions.
- **Trust Algorithms:** Weighted scoring based on verified transactions.
- **Privacy Controls:** Zero-knowledge proofs for selective disclosure.
- **Interoperability:** APIs for marketplaces, DAOs, and social platforms.

## **Benefits**

- Trust without intermediaries.
- Portable reputation across ecosystems.
- Fraud reduction via verifiable records.
- Empowerment of freelancers and communities.

## **Technical Architecture**

- **Layer 1 Anchoring:** Reputation events anchored on Bitcoin.
- **OPNet Smart Contracts:** Handle scoring logic, dispute resolution, and identity binding.

- **Frontend Web App:** User dashboard for managing identity and viewing reputation.
- **Integration APIs:** Allow external platforms to query reputation scores.